# IIMII CONIC SECTIONS

## # General Definition of a Conic

Locus of point $P(x,y)$ whose distance from a fixed point $S$ (Focus) bears a constant ratio $e$ (Eccentricity) to its perpendicular distance from a fixed line $L$ (Directrix).

☆☆

$$\frac{SP}{PM} = e \iff SP^2 = e^2 PM^2$$

Yaad rakho: Agar Focus directrix par lie kare, toh conic Degenerate ho jata hai (Pair of straight lines)!

## # General 2nd Degree Equation & Nature of Conic

General curve: $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$. Discriminant $\Delta$:

☆ $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$

| Condition | Nature of Curve | Eccentricity ($e$) |
|---|---|---|
| $\Delta = 0, h^2 - ab > 0$ | Pair of intersecting lines | — |
| $\Delta = 0, h^2 - ab = 0$ | Pair of parallel lines | — |
| $\Delta \neq 0, a = b, h = 0$ | Circle | $e = 0$ |
| $\Delta \neq 0, h^2 = ab$ | Parabola | $e = 1$ |
| $\Delta \neq 0, h^2 < ab$ | Ellipse | $0 < e < 1$ |
| $\Delta \neq 0, h^2 > ab$ | Hyperbola | $e > 1$ |
| $\Delta \neq 0, h^2 > ab, a + b = 0$ | Rectangular Hyperbola | $e = \sqrt{2}$ |

**Ex.1 :** Classify $x^2 + 4xy + 4y^2 - 6x - 12y + 5 = 0$.

Here $a = 1, h = 2, b = 4 \implies h^2 - ab = 4 - 4 = 0$. Now check $\Delta$:

$\Delta = (1)(4)(5) + 2(-6)(-3)(2) - 1(36) - 4(9) - 5(4) = 20 + 72 - 36 - 36 - 20 = 0$.

Since $\Delta = 0$ and $h^2 = ab$, it represents Parallel lines $(x + 2y - 1)(x + 2y - 5) = 0$.

## # Parabola: Standard Forms & Terminology

Equation: $y^2 = 4ax$ $(a > 0)$. Standard parameters and comparison:

| Feature | $y^2 = 4ax$ | $x^2 = 4ay$ |
|---|---|---|
| Vertex $(V)$ | $(0, 0)$ | $(0, 0)$ |
| Focus $(S)$ | $(a, 0)$ | $(0, a)$ |
| Directrix | $x = -a \iff x + a = 0$ | $y = -a \iff y + a = 0$ |
| Axis of Symmetry | $y = 0$ $(X\text{-axis})$ | $x = 0$ $(Y\text{-axis})$ |
| Tangent at Vertex | $x = 0$ $(Y\text{-axis})$ | $y = 0$ $(X\text{-axis})$ |
| Latus Rectum $(LR)$ | Length $= 4a$, ends $(a, \pm 2a)$ | Length $= 4a$, ends $(\pm 2a, a)$ |
| Focal Distance of $P(x_1, y_1)$ | $SP = x_1 + a$ | $SP = y_1 + a$ |

Dhyan rakho: Agar parabola $(y - k)^2 = 4a(x - h)$ ho toh vertex $(h, k)$ shift ho jata hai!

## # Parametric Coordinates of Parabola

Any point on $y^2 = 4ax$ can be represented in terms of single parameter $t$:

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$$P(t) \equiv (at^2, 2at), \quad t \in \mathbb{R}$$

Here $t = \dfrac{1}{m}$, where $m$ is the slope of tangent at $P$! Slope of tangent is $\dfrac{1}{t}$.

**Ex.2** : Find vertex, focus, and directrix of $y^2 - 4y - 8x + 12 = 0$.

Completing square: $(y - 2)^2 = 8x - 8 = 8(x - 1) = 4(2)(x - 1) \implies a = 2$.

Vertex: $(h, k) = (1, 2)$. Focus: $(h + a, k) = (1 + 2, 2) = (3, 2)$.

Directrix: $x - h = -a \implies x - 1 = -2 \implies \boxed{x + 1 = 0}$.

**Tip** : Length of focal distance of $(at^2, 2at)$ is $SP = a(1 + t^2)$.

Distance between focus $(a, 0)$ and point $(at^2, 2at)$ is $\sqrt{(at^2 - a)^2 + (2at)^2} = a(t^2 + 1)$.

## # Properties of Focal Chord ($y^2 = 4ax$)

Chord passing through Focus $S(a,0)$ with extremities $P(t_1)$ and $Q(t_2)$:

☆☆ Endpoints condition: $\boxed{t_1 t_2 = -1} \implies$ If one end is $t$, other end is $\boxed{-1/t}$.

☆☆ Length of Focal Chord:

$$\boxed{L = a\left(t + \frac{1}{t}\right)^2 = 4a\csc^2\theta}$$

where $\theta$ is inclination of chord with axis. Minimum length is $4a$ (Latus Rectum at $\theta = 90^\circ$).

☆ Semi-latus rectum is Harmonic Mean of focal segments:

$$\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a} = \frac{2}{2a}$$ (Super repeated in JEE Main!)

☆ Circle on any focal chord as diameter touches the directrix. Circle on focal distance $SP$ as diameter touches the tangent at vertex.

## # Position of Point with Respect to Parabola

Let $S(x,y) = y^2 - 4ax$. For point $P(x_1, y_1)$, calculate $S_1 = y_1^2 - 4ax_1$:

$S_1 > 0 \implies$ Point lies Outside the parabola (2 real tangents possible).

$S_1 = 0 \implies$ Point lies On the parabola (1 tangent).

$S_1 < 0 \implies$ Point lies Inside the parabola (No real tangent).

**Ex.3** : If one end of a focal chord of $y^2 = 8x$ is $(2,4)$, find length of the chord.

Here $4a = 8 \implies a = 2$. Point $(2,4) \implies 2at = 4 \implies 2(2)t = 4 \implies t = 1$.

Other end has parameter $t_2 = -1/t = -1$.

Length $= a(t + 1/t)^2 = 2(1+1)^2 = 2(4) = \boxed{8}$.

**Ex.3b** : If $SP = 6$ for $y^2 = 12x$, find $SQ$ for focal chord $PQ$.

Here $4a = 12 \implies a = 3$. Harmonic mean relation: $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a} = \frac{1}{3}$.

$\frac{1}{SQ} = \frac{1}{3} - \frac{1}{6} = \frac{1}{6} \implies SQ = \boxed{6}$. (Here $P, Q$ are ends of $LR$!).

## # Equations of Tangents to Parabola ($y^2 = 4ax$)

**1. Slope Form:** Line $y = mx + c$ is tangent to $y^2 = 4ax$ iff $\boxed{c = \frac{a}{m}}$

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$$\boxed{y = mx + \frac{a}{m} \quad (m \neq 0)}$$

with point of contact $\boxed{\left(\frac{a}{m^2}, \frac{2a}{m}\right)}$

**2. Parametric Form ($t$):** ☆☆ $\boxed{ty = x + at^2}$ (Slope is $1/t$)

**3. Point Form ($x_1, y_1$):** $T = 0 \implies \boxed{yy_1 = 2a(x + x_1)}$

## # Point of Intersection of Tangents at $P(t_1)$ and $Q(t_2)$

Solving $t_1 y = x + at_1^2$ and $t_2 y = x + at_2^2$:

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$$\boxed{T(h,k) = \Big(at_1t_2,\ a(t_1 + t_2)\Big)}$$

$x$-coord is G.M. of abscissae; $y$-coord is A.M. of ordinates of $P$ and $Q$!

## # Director Circle of Parabola

Locus of point of intersection of two mutually perpendicular tangents ($m_1m_2 = -1$).

From $T(at_1t_2, a(t_1 + t_2))$, perpendicular tangents $\implies t_1t_2 = -1 \implies x = a(-1) = -a$.

☆☆ Director circle of a parabola is its $\boxed{\text{Directrix: } x + a = 0}$!

Yaad rakho: Parabola ka director circle circle nahi hota, ek straight line (directrix) hoti hai!

**Ex.4 :** Find common tangent to $y^2 = 4x$ and $x^2 = -32y$.

Tangent to $y^2 = 4x$ ($a = 1$): $y = mx + 1/m$. Substitute into $x^2 = -32y$:

$x^2 + 32mx + 32/m = 0$. For tangency: $D = 0 \implies (32m)^2 - 4(32/m) = 0 \implies m^3 = 1/8 \implies m = 1/2$.

Tangent: $y = \frac{1}{2}x + 2 \implies \boxed{x - 2y + 4 = 0}$.

## # Equations of Normals to Parabola ($y^2 = 4ax$)

**1. Parametric Form ($t$):** ☆☆ $\boxed{y + tx = 2at + at^3}$ *(Slope is $m = -t$)*

**2. Slope Form ($m$):** Put $t = -m$:

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$$\boxed{y = mx - 2am - am^3}$$

with foot of normal $\boxed{(am^2, -2am)}$

**3. Point Form ($x_1, y_1$):** $y - y_1 = -\frac{y_1}{2a}(x - x_1)$

## # Co-Normal Points (Three Normals from $(h, k)$)

If normal passes through $(h, k)$: $k = mh - 2am - am^3 \implies am^3 + (2a - h)m + k = 0$.

☆ Roots $m_1, m_2, m_3$ satisfy:

$\sum m_i = m_1 + m_2 + m_3 = 0$ *(No $m^2$ term in cubic!)*

$\sum m_1 m_2 = \frac{2a - h}{a}, \quad m_1 m_2 m_3 = -\frac{k}{a}$

☆☆ Sum of ordinates of co-normal points: $\boxed{\sum y_i = -2a \sum m_i = 0}$

☆ Centroid of $\triangle$ formed by co-normal points: $\boxed{G = \left(\frac{2h - 4a}{3}, 0\right)}$ *(Always on axis!)*

☆ Condition for three distinct real normals: $\boxed{h > 2a}$ and $27ak^2 < 4(h - 2a)^3$.

## # Relation Between Parameters if Normal at $t_1$ meets at $t_2$

If the normal at $P(t_1)$ intersects the parabola again at $Q(t_2)$:

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$$\boxed{t_2 = -t_1 - \frac{2}{t_1}}$$

*Minimum length of such a normal chord is $6\sqrt{3}a$ (at $t_1 = \sqrt{2}$).*

## # Chords of Parabola ($y^2 = 4ax$)

**1. Chord joining $P(t_1)$ and $Q(t_2)$:**

☆ Slope $= \frac{2}{t_1 + t_2}$. Equation: $\boxed{2x - (t_1 + t_2)y + 2at_1t_2 = 0}$

**2. Chord of Contact ($T = 0$):** From external point $(x_1, y_1)$

☆☆ Equation: $\boxed{yy_1 = 2a(x + x_1)}$. Area of $\triangle$ formed by tangents & chord: $\frac{(y_1^2 - 4ax_1)^{3/2}}{2a}$.

**3. Chord with Given Midpoint $(x_1, y_1)$:** ☆☆ $\boxed{T = S_1}$

$yy_1 - 2a(x + x_1) = y_1^2 - 4ax_1 \implies \boxed{yy_1 - 2ax = y_1^2 - 2ax_1}$

## # Important Geometric & Reflection Properties

☆ *Reflection Property:* Any ray parallel to the axis of parabola, after reflection from the inside surface, passes through the Focus $S$.

☆ Tangent at any point $P$ bisects the angle between focal radius $SP$ and perpendicular from $P$ to directrix.

☆ Portion of any tangent intercepted between point of contact and directrix subtends a right angle ($90°$) at the focus.

☆ The locus of foot of perpendicular from focus upon any tangent is the $\boxed{\text{Tangent at vertex } (x = 0)}$.

**Ex.5 :** Find midpoint chord of $y^2 = 8x$ whose midpoint is $(3, 2)$.

Here $a = 2$. Using $T = S_1$: $y(2) - 2(2)(x + 3) = (2)^2 - 8(3)$.

$2y - 4x - 12 = 4 - 24 = -20 \implies 2y - 4x + 8 = 0 \implies \boxed{2x - y - 4 = 0}$.

**Tip :** Subtangent and Subnormal for $y^2 = 4ax$:

Length of subtangent $= 2x_1$ (bisected at vertex). Length of subnormal $= 2a$ (constant for any point!).

## # Ellipse: Standard Equations & Terminology

Equation: $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ $(0<e<1)$. Two standard orientations:

| Feature | Horizontal $(a>b)$ | Vertical $(b>a)$ |
|---|---|---|
| Eccentricity $(e)$ | $b^2=a^2(1-e^2) \implies e=\sqrt{1-b^2/a^2}$ | $a^2=b^2(1-e^2) \implies e=\sqrt{1-a^2/b^2}$ |
| Foci $(S, S')$ | $(\pm ae, 0)$ | $(0, \pm be)$ |
| Directrices | $x=\pm\frac{a}{e}$ | $y=\pm\frac{b}{e}$ |
| Latus Rectum ( $LR$) | $\frac{2b^2}{a}$ | $\frac{2a^2}{b}$ |
| Major & Minor Axes | $2a$ $(X\text{-axis})$, $2b$ $(Y\text{-axis})$ | $2b$ $(Y\text{-axis})$, $2a$ $(X\text{-axis})$ |
| Focal Distance Sum | $\boxed{SP+S'P=2a}$ (Const.) | $\boxed{SP+S'P=2b}$ |

Dhyan rakho: $SP+S'P=2a$ is mechanical definition used to draw ellipse using two pins and a loop of string!

## # Auxiliary Circle & Parametric Representation

Auxiliary circle: Circle described on major axis as diameter: $\boxed{x^2+y^2=a^2}$.

Let $Q(a\cos\theta, a\sin\theta)$ be on auxiliary circle, then corresponding point $P$ on ellipse is:

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$$\boxed{P(\theta)\equiv(a\cos\theta,\ b\sin\theta),\quad \theta\in[0,2\pi)}$$

Angle $\theta$ is Eccentric Angle (NOT polar angle of $P$, it is the angle made by $OQ$ with $X$-axis!).

**Ex.6** : Find eccentricity, foci, and latus rectum of $9x^2+16y^2=144$.

Dividing by 144: $\frac{x^2}{16}+\frac{y^2}{9}=1 \implies a^2=16, b^2=9 \implies a=4, b=3$ $(a>b)$.

$e=\sqrt{1-9/16}=\frac{\sqrt{7}}{4} \implies$ Foci: $(\pm ae,0)=\boxed{(\pm\sqrt{7},0)}$.

Length of $LR=\frac{2b^2}{a}=\frac{2(9)}{4}=\boxed{\frac{9}{2}}$.

## # Line and Ellipse: Condition of Tangency

Line $y = mx + c$ is tangent to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ iff:

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$$c^2 = a^2m^2 + b^2 \implies c = \pm\sqrt{a^2m^2 + b^2}$$

Point of contact: $\left(-\frac{a^2m}{c}, \frac{b^2}{c}\right)$ where $c = \pm\sqrt{a^2m^2 + b^2}$.

## # Forms of Tangents to Ellipse

1. Slope Form: ☆☆ $y = mx \pm \sqrt{a^2m^2 + b^2}$

2. Parametric Form ($\theta$): ☆☆ $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$

3. Point Form $(x_1, y_1)$: $T = 0 \implies \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$

## # Director Circle of Ellipse

Locus of point of intersection of two mutually perpendicular tangents to the ellipse:

☆☆

$$x^2 + y^2 = a^2 + b^2$$

Concentric circle with center $(0, 0)$ and radius $\sqrt{a^2 + b^2}$.

## # Important Tangent Properties

☆ Product of Perpendiculars from foci $S, S'$ onto ANY tangent is constant:

$p_1 \cdot p_2 = b^2$ !

☆ Feet of perpendiculars from foci onto any tangent lie on

Auxiliary Circle: $x^2 + y^2 = a^2$.

**Ex.7** : Find tangents to $x^2 + 4y^2 = 4$ having slope $m = 1$.

Standard form: $\frac{x^2}{4} + \frac{y^2}{1} = 1 \implies a^2 = 4, b^2 = 1$. With $m = 1$:

$c = \pm\sqrt{a^2m^2 + b^2} = \pm\sqrt{4(1)^2 + 1} = \pm\sqrt{5}$.

Tangents: $y = x \pm \sqrt{5} \implies x - y \pm \sqrt{5} = 0$.

## # Equations of Normals to Ellipse

1. Point Form $(x_1, y_1)$: ☆☆ $\boxed{\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2}$

2. Parametric Form $(\theta)$: ☆☆ $\boxed{ax\sec\theta - by\csc\theta = a^2 - b^2}$

3. Slope Form $(m)$: $y = mx - \frac{m(a^2 - b^2)}{\sqrt{a^2 + b^2m^2}}$

*At most 4 normals can be drawn from an external point to an ellipse.*

## # Chords of Ellipse

1. Chord joining $\theta_1$ and $\theta_2$:

☆ $\frac{x}{a}\cos\left(\frac{\theta_1 + \theta_2}{2}\right) + \frac{y}{b}\sin\left(\frac{\theta_1 + \theta_2}{2}\right) = \cos\left(\frac{\theta_1 - \theta_2}{2}\right)$

*If this chord passes through focus* $(ae, 0)$, *then:* $\tan(\theta_1/2)\cdot\tan(\theta_2/2) = \frac{e-1}{e+1}$.

2. Chord of Contact $(T = 0)$: $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$

3. Chord with Midpoint $(x_1, y_1)$: ☆☆ $\boxed{T = S_1} \implies \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$

## # Reflection Property of Ellipse

A ray emanating from one focus $S$, after reflection from the inner surface of the ellipse, passes through the other focus $S'$.

*Normal at point* $P$ *is internal bisector of* $\angle SPS'$, *and tangent is external bisector!*

**Ex.8** : Find equation of normal to $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at $(2, 3\sqrt{3}/2)$.

Here $\theta = \pi/3$. Using parametric form: $ax\sec\theta - by\csc\theta = a^2 - b^2$:

$4x\sec(\pi/3) - 3y\csc(\pi/3) = 16 - 9 = 7 \implies 4x(2) - 3y(2/\sqrt{3}) = 7.$

$8x - 2\sqrt{3}y - 7 = 0$

## # Hyperbola: Standard Equation & Terminology

Standard equation: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ $(e > 1)$.

| Feature | Standard: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ | Conjugate: $-\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ |
|---|---|---|
| Eccentricity $(e)$ | $b^2 = a^2(e^2 - 1) \implies e = \sqrt{1 + b^2/a^2}$ | $a^2 = b^2(e'^2 - 1) \implies e' = \sqrt{1 + a^2/b^2}$ |
| Foci $(S, S')$ | $(\pm ae, 0)$ | $(0, \pm be')$ |
| Directrices | $x = \pm\frac{a}{e}$ | $y = \pm\frac{b}{e'}$ |
| Latus Rectum ( $LR$) | $\frac{2b^2}{a}$ | $\frac{2a^2}{b}$ |
| Focal Distance Diff. | $\boxed{\lvert SP - S'P\rvert = 2a}$ | $\boxed{\lvert SP - S'P\rvert = 2b}$ |

## # Relation Between Eccentricities of Conjugate Hyperbolas

If $e$ and $e'$ are eccentricities of a hyperbola and its conjugate hyperbola:

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$$\boxed{\frac{1}{e^2} + \frac{1}{e'^2} = 1}$$

Super frequent direct JEE formula! If $e = 2$, then $1/e'^2 = 1 - 1/4 = 3/4 \implies e' = 2/\sqrt{3}$.

## # Parametric Coordinates of Hyperbola

Point on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$: $\boxed{P(\theta) \equiv (a\sec\theta,\ b\tan\theta)}$, $\theta \in [0, 2\pi) \setminus \{\pi/2, 3\pi/2\}$.

**Ex.9 :** If eccentricity of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $e = 5/4$, find eccentricity of conjugate.

Using $\frac{1}{e^2} + \frac{1}{e'^2} = 1 \implies \frac{16}{25} + \frac{1}{e'^2} = 1 \implies \frac{1}{e'^2} = \frac{9}{25}$.

Thus $e'^2 = \frac{25}{9} \implies e' = \boxed{\frac{5}{3}}$.

## # Tangents to Hyperbola ($dfracx^2a^2 - dfracy^2b^2 = 1$)

**1. Condition of Tangency:** Line $y = mx + c$ is tangent iff $\boxed{c^2 = a^2m^2 - b^2}$

*Requires $|m| > b/a$ for real tangents! If $|m| = b/a$, line is parallel to asymptotes.*

**2. Slope Form:** ☆☆ $\boxed{y = mx \pm \sqrt{a^2m^2 - b^2}}$

**3. Parametric Form ($heta$):** ☆☆ $\boxed{\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1}$

**4. Point Form $(x_1, y_1)$:** $T = 0 \implies \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

## # Director Circle of Hyperbola

Locus of point of intersection of two perpendicular tangents:

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$$\boxed{x^2 + y^2 = a^2 - b^2}$$

If $a > b$: Real circle with radius $\sqrt{a^2 - b^2}$.

If $a = b$: Point circle $(0, 0)$ (tangents only intersect perpendicularly at origin).

If $a < b$: No real perpendicular tangents exist (imaginary director circle).

## # Normals to Hyperbola

**1. Point Form $(x_1, y_1)$:** ☆☆ $\boxed{\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2}$

**2. Parametric Form ($heta$):** ☆☆ $\boxed{ax\cos\theta + by\cot\theta = a^2 + b^2}$

**3. Slope Form ($m$):** $y = mx \pm \frac{m(a^2 + b^2)}{\sqrt{a^2 - b^2m^2}}$

**Ex.9b :** Tangent to $\frac{x^2}{9} - \frac{y^2}{4} = 1$ with slope $m = 2$:

$c = \pm\sqrt{a^2m^2 - b^2} = \pm\sqrt{9(4) - 4} = \pm\sqrt{32} = \pm4\sqrt{2}$. Tangent: $y = 2x \pm 4\sqrt{2}$.

## # Asymptotes of Hyperbola

Straight lines passing through center and tangent to hyperbola at infinity:

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$$y = \pm\frac{b}{a}x \iff \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$

☆ Angle between asymptotes: $2\theta = 2\tan^{-1}(b/a) = 2\sec^{-1}(e)$.

☆ Differs by Constant Rule: For any hyperbola $H$, its asymptotes $A$, and conjugate $C$:

$H + C = 2A \implies (H - A) + (C - A) = 0$ (They differ only in constant term!)

☆ Product of perpendiculars from ANY point on hyperbola to asymptotes: $\frac{a^2b^2}{a^2+b^2}$.

## # Rectangular Hyperbola ($a = b \iff e = \sqrt{2}$)

Asymptotes are perpendicular ($x^2 - y^2 = 0 \implies y = \pm x$).

**Rotated Form:** $xy = c^2$ (Axes rotated by $45°$, $c^2 = a^2/2$)

☆ Parametric point: $P(t) \equiv \left(ct, \frac{c}{t}\right)$, $t \neq 0$.

☆ Tangent at $t$: $x + yt^2 = 2ct \iff \frac{x}{t} + yt = 2c$.

Intercepts on axes: $A(2ct, 0)$ and $B(0, 2c/t)$. Area of $\triangle OAB = \frac{1}{2}(2ct)(2c/t) = 2c^2$ (Constant!). Point $P$ bisects $AB$!

☆ Normal at $t$: $xt^3 - yt = ct^4 - c$.

**Ex.10 :** If a triangle has vertices on $xy = c^2$, where does its orthocenter lie?

Let vertices be $t_1, t_2, t_3$. Orthocenter is $H\left(-\frac{c}{t_1t_2t_3}, -ct_1t_2t_3\right)$.

Notice: $x_H \cdot y_H = (-c/T)(-cT) = c^2 \implies$ Orthocenter also lies on the hyperbola!

## # Common Tangents Across Different Conics

Systematic recipe to find common tangent between two curves:

1. Write slope-form tangent of Conic 1: e.g. for $y^2 = 4ax \implies y = mx + a/m$.
2. Apply condition of tangency for Conic 2:

$\rightarrow$ For circle $x^2 + y^2 = r^2$: perpendicular distance from center $= r \implies \dfrac{|a/m|}{\sqrt{1+m^2}} = r$

.

$\rightarrow$ For ellipse $x^2/A^2 + y^2/B^2 = 1$: equate $c^2 \implies (a/m)^2 = A^2m^2 + B^2$.

3. Solve the polynomial for $m$, then write the lines.

## # Pole and Polar

For point $P(x_1, y_1)$ (Pole), its Polar with respect to any conic $S = 0$ is $\boxed{T = 0}$.

- ☆ If $P$ is outside, Polar is the Chord of Contact.
- ☆ If $P$ is on the curve, Polar is the Tangent.
- ☆ Polar of Focus is the Directrix!
- ☆ Conjugate Points: If polar of $P$ passes through $Q$, then polar of $Q$ passes through $P$.

## # Summary of Standard Eccentricities & Director Circles

| Conic | Equation | Eccentricity ($e$) | Director Circle |
|---|---|---|---|
| Parabola | $y^2 = 4ax$ | $e = 1$ | $x + a = 0$ (Directrix) |
| Ellipse | $x^2/a^2 + y^2/b^2 = 1$ | $\sqrt{1 - b^2/a^2} < 1$ | $x^2 + y^2 = a^2 + b^2$ |
| Hyperbola | $x^2/a^2 - y^2/b^2 = 1$ | $\sqrt{1 + b^2/a^2} > 1$ | $x^2 + y^2 = a^2 - b^2$ $(a > b)$ |
| Rect. Hyp. | $xy = c^2$ | $e = \sqrt{2}$ | Point $(0,0)$ (origin) |

## # Good Ques. (JEE Main Style)

**Q1.** For parabola $y^2 = 8x$, find the length of the focal chord having one extremity at $(2, 4)$.

**Ans : 8**

**Q2.** Find the locus of point of intersection of two perpendicular tangents to parabola $y^2 = 12x$.

**Ans : $x + 3 = 0$**

**Q3.** For ellipse $\frac{x^2}{16} + \frac{y^2}{7} = 1$, find the distance between its two foci.

**Ans : 6**

**Q4.** Find area of rectangle formed by ends of latera recta of ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

**Ans : $\frac{144}{5}$**

**Q5.** If eccentricity of hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$ is $e$, find eccentricity of its conjugate.

**Ans : $\frac{5}{4}$**

**Q6.** Find the common tangent to parabola $y^2 = 4x$ and parabola $x^2 = -32y$.

**Ans : $x - 2y + 4 = 0$**

**Q7.** Find equation of normal to $y^2 = 4x$ at point $(4, 4)$.

**Ans : $2x + y = 12$**

**Q8.** What is the eccentricity of the rectangular hyperbola $xy = 16$?

**Ans : $\sqrt{2}$**

## # Good Ques. (JEE Adv Style)

**Q1.** Normals at 3 points on $y^2 = 4ax$ meet at $(h, k)$. Find centroid of triangle formed by the feet.

**Ans : $\left(\frac{2h - 4a}{3}, 0\right)$**

**Q2.** Find common tangent to ellipses $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and $\frac{x^2}{9} + \frac{y^2}{16} = 1$ in Ist quadrant.

**Ans : $x + y = 5$**

**Q3.** An ellipse has foci at $(0, 2)$ and $(0, -2)$ and length of minor axis is 4. Find length of major axis.

**Ans : $4\sqrt{2}$**

**Q4.** Find the length of the common chord of parabola $y^2 = 4x$ and circle $x^2 + y^2 = 12$.

**Ans : $4\sqrt{2}$**

**Q5.** Find product of perpendiculars from any point on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ to its asymptotes.

**Ans : $\frac{a^2b^2}{a^2 + b^2}$**

**Q6.** Tangent at $P(ct, c/t)$ to $xy = c^2$ meets axes at $A$ and $B$. Find area of $\triangle OAB$.

**Ans : $2c^2$**

## # Kaunsa Method Lagaun ?

| Jab question me yeh dikhe... | Toh direct yeh method lagao ! |
|---|---|
| Common tangent between 2 conics | $\Rightarrow$ Slope form $y = mx + c$ likho aur dusre conic ki tangency condition se $m$ nikaalo. |
| Chord with given midpoint $(x_1, y_1)$ | $\Rightarrow$ Seedha $\boxed{T = S_1}$ lagao (Parabola, Ellipse, Hyperbola sabke liye valid!). |
| Focal chord property / length | $\Rightarrow$ Parametric formula $t_1 t_2 = -1$ aur length $= a(t + 1/t)^2 = 4a\,\mathrm{cosec}^2\,\theta$. |
| Locus of intersection of $\perp$ tangents | $\Rightarrow$ Director circle: Parabola $\to x + a = 0$, Ellipse $\to x^2 + y^2 = a^2 + b^2$, Hyp $\to x^2 + y^2 = a^2 - b^2$. |
| Normals from point $(h, k)$ to parabola | $\Rightarrow$ Cubic $am^3 + (2a - h)m + k = 0$ banao, $\sum m_i = 0$ aur $\sum y_i = 0$ use karo. |

## # Last Minute Formula Strip

**Parabola Tangent**

$y = mx + \dfrac{a}{m}$

$P = (a/m^2, 2a/m)$

**Parabola Normal**

$y = mx - 2am - am^3$

Foot $= (am^2, -2am)$

**Ellipse Tangent**

$y = mx \pm \sqrt{a^2m^2 + b^2}$

Director: $x^2 + y^2 = a^2 + b^2$

**Hyperbola Tangent**

$y = mx \pm \sqrt{a^2m^2 - b^2}$

Director: $x^2 + y^2 = a^2 - b^2$

**Conjugate Hyp.** $e$

$\dfrac{1}{e^2} + \dfrac{1}{e'^2} = 1$

Direct formula!

**Rect. Hyp.** $xy = c^2$

$x + yt^2 = 2ct$

Area $\triangle OAB = 2c^2$

## # Dhyan Rakho ! (Common Galtiyan)

- ☆ **Vertical Parabola / Ellipse Trap:** Parabola $x^2 = 4ay$ me tangent $y = mx - am^2$ hoti hai, $y = mx + a/m$ mat laga dena!
- ☆ **Hyperbola Tangent Slope Condition:** Hyperbola me real tangent ke liye $|m| > b/a$ hona zaroori hai. Agar $|m| \le b/a$, koi tangent exist nahi karti!
- ☆ **Focal Distance Definition:** Ellipse me sum $SP + S'P = 2a$ hota hai, jabki Hyperbola me difference $|SP - S'P| = 2a$ hota hai.
- ☆ **Director Circle of Hyperbola:** Agar $a < b$ ho, toh hyperbola ka koi real director circle nahi hota (radius $\sqrt{a^2 - b^2}$ imaginary ho jati hai)!

## # Hints (Separately for JEE Main Questions)

**Q1.** $a = 2, 2at = 4 \implies t = 1$. Use $L = a(t + 1/t)^2 = 2(1 + 1)^2 = 8$.

**Q2.** Locus of perpendicular tangents is directrix $x = -a$. Here $4a = 12 \implies a = 3 \implies x + 3 = 0$.

**Q3.** $a^2 = 16, b^2 = 7 \implies e = \sqrt{1 - 7/16} = 3/4$. Distance between foci $= 2ae = 2(4)(3/4) = 6$.

**Q4.** Ends of latera recta are $(\pm ae, \pm b^2/a)$. Area $= 2ae \times 2b^2/a = 8 \times 18/5 = 144/5$.

**Q5.** $e^2 = 1 + 16/9 = 25/9$. Use $1/e^2 + 1/e'^2 = 1 \implies 1/e'^2 = 16/25 \implies e' = 5/4$.

**Q6.** Tangent $y = mx + 1/m$. Put in $x^2 = -32y$, $D = 0 \implies m = 1/2 \implies x - 2y + 4 = 0$.

**Q7.** Here $a = 1, t = 2$. Normal is $y + tx = 2at + at^3 \implies y + 2x = 4 + 8 = 12$.

**Q8.** In rectangular hyperbola asymptotes are perpendicular $\implies e = \sqrt{2}$.

## # Hints (Separately for JEE Advanced Questions)

**Q1.** Feet are $(am_i^2, -2am_i)$. $\sum m_i = 0 \implies y_G = 0$. $x_G = \frac{a}{3}(\sum m_i^2) = \frac{a}{3}(-2\sum m_1 m_2) = \frac{2h-4a}{3}$.

**Q2.** For ellipse 1: $c^2 = 16m^2 + 9$. For ellipse 2: $c^2 = 9m^2 + 16$. Equating gives $m = -1, c = 5 \implies x + y = 5$.

**Q3.** Foci on $Y$-axis $\implies be = 2, a = 2$. $a^2 = b^2 - (be)^2 \implies 4 = b^2 - 4 \implies b = 2\sqrt{2} \implies 2b = 4\sqrt{2}$.

**Q4.** Put $y^2 = 4x$ into $x^2 + 4x - 12 = 0 \implies x = 2 \implies y = \pm 2\sqrt{2}$. Chord length $= 4\sqrt{2}$.

**Q5.** Asymptotes are $bx \pm ay = 0$. Product of distances is $\frac{|b^2x^2 - a^2y^2|}{b^2 + a^2} = \frac{a^2b^2}{a^2 + b^2}$.

**Q6.** Tangent $x + yt^2 = 2ct$. Intercepts are $A(2ct, 0), B(0, 2c/t)$. Area $= \frac{1}{2}(2ct)(2c/t) = 2c^2$.

## # Advanced Me Kya Dhyan Rakhein

☆☆ *Parametric form over Cartesian:* Variable tangents or chords me parameter $t$ or $\theta$ use karo. Calculations $3\times$ fast ho jaati hain!

☆☆ *Homogenization Shortcut:* Chord subtending $90^\circ$ at origin? Homogenize conic with line $lx + my = 1$ and set coeff($x^2$) + coeff($y^2$) $= 0$.

☆☆ *Auxiliary Circle & Projection:* Ellipse area/tangent problems ko auxiliary circle par project karke solve karo ($\text{Area}_{\text{ellipse}} = \frac{b}{a} \times \text{Area}_{\text{circle}}$).